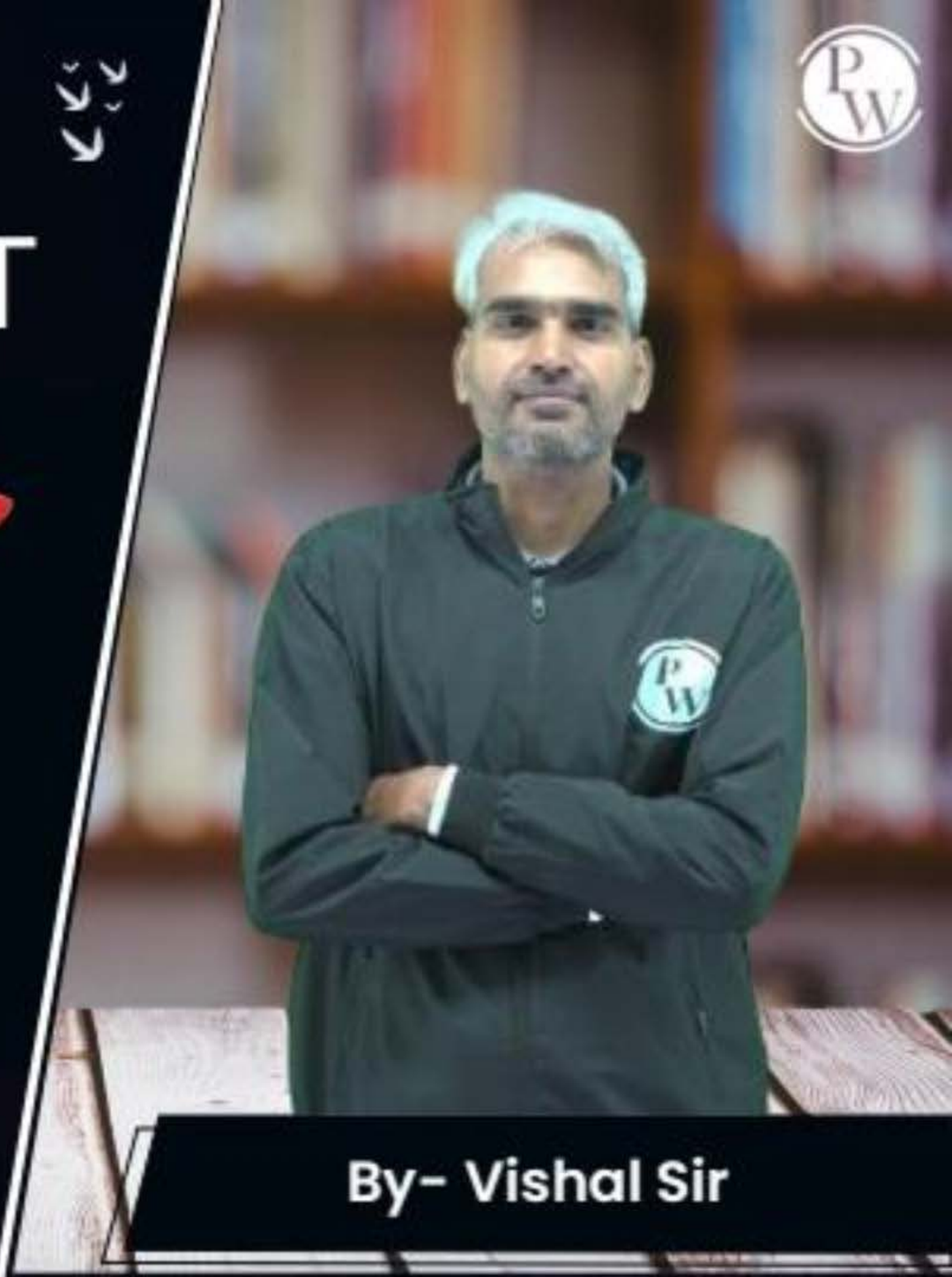
Computer Science & IT



Compiler Design

Syntax Analysis (Parser)

Lecture No. 05



By- Vishal Sir

Recap of Previous Lecture



- Topic: Elimination of Left Recursion
- Topic: Left Factoring
- Topic: FIRST set computation

Slide

Topics to be Covered



Topic FOLLOW set computation

Topic Questions on FIRST & FOLLOW

Slide



Topic : First



First of X, is Set of all terminals that may begin the string derived from X.

① if X is a terminal
First(x) = {x}

② If X is a non-terminal, and X → ABC is a production in the gramman, then
First(X) = First(ABC)

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if X → ABC / PQR / LMN
First(X) = First(ABC) U First(PQR) U First(LMN)

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Topic : Follow

* Follow of X is the set of all terminals that can appear immediately on the right-side of non-terminal X, in some sentential form.

* Follow set is computed only for non-terminals of the grammar.

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Topic : Follow

$S \to \alpha_1 A \beta_1$

* Process to Compute Follow Set of non-terminal 'X'

① If X is the start symbol, then add $ into Follow(X).

② If $A \to \alpha X \beta$ is the production in the grammar, then

(i) if $\epsilon \notin$ first(β), then add first(β) into follow(X)

(ii) if $\epsilon \in$ first(β), then add [(first(β) − {ϵ}) ∪ follow(A)] into follow(X,

Union

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Topic : Follow

③ If $A \longrightarrow \alpha X$ is the production in the grammar, then add follow(A) into follow(X)

Note: follow Set never Contains 'ϵ'.

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Topic : First & Follow

	Follow
$S \rightarrow aAB$	{ $ }
A → b	
B → c	

$S \rightarrow \underbrace{a}_{\alpha} A \underbrace{B}_{\beta} \Rightarrow follow(A) = first(B) = \{ c \}$

$S \rightarrow \underbrace{aAB}_{\alpha} \Rightarrow follow(B) = follow(S) = \{ \$ \}$

Slide



Topic : First & Follow

	first	Follow
$S \rightarrow AB$	{a, b}	{$} only bcoz 'S' is not present R.H.S. of any production
$A \rightarrow a / \epsilon$	{a, ∈}	$S \rightarrow AB$ follow(A) = first(B) = {b}
$B \rightarrow b$	{b}	$S \rightarrow AB$ follow(B) = follow(S) = {$}

Slide



Topic : First & Follow

Start symbol	first	Follow
X→aABe	{a b	{$}
B→c/d	{c,d}	$X \to \underset{\alpha}{a}A\underset{\beta}{Be} \Rightarrow follow(B) = first(e) = \{e\}$
A→a	{a}	$X \to \underset{\alpha}{a}\underset{\beta}{ABe} \Rightarrow follow(A) = first(Be) = \{c,d\}$

Slide



Topic : First & Follow

	First	Follow
$E \to TE'$	{id, (}	{$,)}
$E' \to +TE'/\varepsilon$	{+, ∈}	{$,)}
$T \to FT'$	{id, (}	
$T' \to *FT'/\varepsilon$	{*, ∈}	
$F \to id/(E)$	{id, (}	

Process

E = start symbol, ∴ Follow(E) = {$}

F → (E), ∴ follow(E) = first( ) = {) }

E → TE', ∴ follow(E') = Follow(E) = { $,) }

E' → +TE', follow(E') = Follow(E')

no change bcoz of this

H.W.

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Topic : First & Follow

S→aBDh

B→cC

C→bc / ε

D→EF

E→g/ ε

F→f/ ε

Slide



Topic : First & Follow

$S \rightarrow AaAb/BbBa$

$A \rightarrow c/\ \varepsilon$

$B \rightarrow d/\ \varepsilon$

Slide



Topic : First & Follow

S→(L)/a

L→SL'

L'→ , SL'/ ε



Topic : First & Follow

S→ACB/ CbB /Ba

A→da/BC

B→g/ ε

C→h/ ε



2 mins Summary

Topic Elimination of Left Recursion

Topic Left Factoring

Topic First & Follow

THANK - YOU